

# Foundations of Adversarial examples

[Adversarial examples]

## 1.0 Content Block

$$\text{adv}_x = x + \epsilon \cdot \text{sign}(\Delta x J(\theta, x, y))$$

## 2.0 Content Block

Concept drift Because malware creators continuously adapt their techniques, the statistical properties of malicious samples also change. This form of concept drift has been widely documented and may reduce model performance unless systems are updated regularly or incorporate mechanisms for incremental learning.

## 3.0 Content Block

$$S(x') := \begin{cases} \max_{c \neq C(x)} F(x')_c - F(x')_{C(x)}, & (\text{Untargeted}) \\ F(x')_{c^*} - \max_{c \neq c^*} F(x')_c, & (\text{Targeted}) \end{cases}$$

## 4.0 Content Block

Model extraction Model extraction involves an adversary probing a black box machine learning system in order to extract the data it was trained on. This can cause issues when either the training data or the model itself is sensitive and confidential. For example, model extraction could be used to extract a proprietary stock trading model which the adversary could then use for their own financial benefit. In the extreme case, model extraction can lead to model stealing, which corresponds to extracting a sufficient amount of data from the model to enable the complete reconstruction of the model. On the other hand, membership inference is a targeted model extraction attack, which infers the owner of a data point, often by leveraging the overfitting resulting from poor machine learning practices. Concerningly, this is sometimes achievable even without knowledge or access to a target model's parameters, raising security concerns for models trained on sensitive data, including but not limited to medical records and/or personally identifiable information. With the emergence of transfer learning and public accessibility of many state of the art machine learning models, tech companies are increasingly drawn to create models based on public ones, giving attackers freely accessible information to the structure and type of model being used.

## 5.0 Content Block

$$\arg \max_{k=1, \dots, K} f_k(\hat{x}) \neq y, \quad ||\hat{x} - x||_p \leq \epsilon \text{ and } \hat{x} \in [0, 1]^d$$

## 6.0 Content Block

Adversarial attacks and training in linear models There is a growing literature about adversarial attacks in linear models. Indeed, since the seminal work from Goodfellow et al. studying these models in linear models has been an important tool to understand how adversarial attacks affect machine learning models. The analysis of these models is simplified because the computation of adversarial attacks can be simplified in linear regression and classification problems. Moreover, adversarial training is convex in this case. Linear models allow for analytical analysis while still reproducing phenomena observed in

state-of-the-art models. One prime example of that is how this model can be used to explain the trade-off between robustness and accuracy. Diverse work indeed provides analysis of adversarial attacks in linear models, including asymptotic analysis for classification and for linear regression. And, finite-sample analysis based on Rademacher complexity. A result from studying adversarial attacks in linear models is that it closely relates to regularization. Under certain conditions, it has been shown that

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$\min_{\hat{x} \in [0, 1]^d} L(f(\hat{x}), y), \text{ s.t. } \|\hat{x} - x\|_p \leq \epsilon$   $\{\displaystyle \min_{\hat{x} \in [0,1]^d} L(f(\hat{x}), y), \text{ s.t. } \|\hat{x} - x\|_p \leq \epsilon\}$

### 8.0 Content Block

Here the objective is to minimize the noise ( $\delta$ ), added to the original input  $x$ , such that the machine learning algorithm ( $C$ ) predicts the original input with  $\delta$  (or  $x + \delta$ ) as some other class  $t$ . However instead of directly the above equation, Carlini and Wagner propose using a new function  $f$  such that:

### 9.0 Content Block

Challenges in applying machine learning to security Researchers have observed that the constraints under which machine-learning techniques function in the security domain are different from those of common benchmark domains. Security data may change over time, include mislabeled samples, or reflect adversarial behavior, which complicates evaluation and reproducibility.

### 10.0 Content Block

When solved using gradient descent, this equation is able to produce stronger adversarial examples when compared to fast gradient sign method that is also able to bypass defensive distillation, a defense that was once proposed to be effective against adversarial examples.

### 11.0 Content Block

Simple black-box adversarial attacks Simple black-box adversarial attacks is a query-efficient way to attack black-box image classifiers. Take a random orthonormal basis  $v_1, v_2, \dots, v_d$  in  $\mathbb{R}^d$ . The authors suggested the discrete cosine transform of the standard basis (the pixels). For a correctly classified image  $x$ , try  $x + \epsilon v_1$ ,  $x - \epsilon v_1$ , and compare the amount of error in the classifier upon  $x + \epsilon v_1$ ,  $x - \epsilon v_1$ . Pick the one that causes the largest amount of error.

### 12.0 Content Block

History At the MIT Spam Conference in January 2004, John Graham-Cumming showed that a machine-learning spam filter could be used to defeat another machine-learning spam filter by automatically learning which words to add to a spam email to get the email classified as not spam. In 2004, Nilesch Dalvi and others noted that linear classifiers used in spam filters could be defeated by simple "evasion attacks" as spammers inserted "good words" into their spam emails. (Around 2007, some spammers added random noise to fuzz words within "image spam" in order to defeat OCR-based filters.) In 2006, Marco Barreno and others published "Can Machine Learning Be Secure?", outlining a broad taxonomy of attacks. As late as 2013 many researchers continued to hope that non-linear classifiers (such as support vector machines and neural networks) might be robust to adversaries, until Battista Biggio and others demonstrated the first gradient-based attacks on such machine-learning models (2012–2013). In 2012, deep neural networks began to dominate computer vision problems; starting in 2014, Christian Szegedy and others demonstrated that deep neural networks could be fooled by adversaries, again using a gradient-based attack to craft adversarial perturbations. Further work would show that adversarial

attacks are harder to produce in uncontrolled environments, due to the different environmental constraints that cancel out the effect of noise. For example, any small rotation or slight illumination on an adversarial image can destroy the adversariality. In addition, researchers such as Google Brain's Nick Frosst point out that it is much easier to make self-driving cars miss stop signs by physically removing the sign itself, rather than creating adversarial examples. Frosst also believes that the adversarial machine learning community incorrectly assumes models trained on a certain data distribution will also perform well on a completely different data distribution. He suggests that a new approach to machine learning should be explored, and is currently working on a unique neural network that has characteristics more similar to human perception than state-of-the-art approaches. While adversarial machine learning continues to be heavily rooted in academia, large tech companies such as Google, Microsoft, and IBM have begun curating documentation and open source code bases to allow others to concretely assess the robustness of machine learning models and minimize the risk of adversarial attacks.